



TECHNICAL DATA SHEET

Square Hollow Section (SHS) — 200 mm × 200 mm × 3 mm

Description: Cold-formed or hot-finished structural steel square tube with a nominal profile of 200 mm by 200 mm and a uniform 3 mm wall thickness. Widely utilized in structural frameworks, machine construction, civil engineering, and general architecture due to its symmetrical strength distribution.

1. Dimensional & Geometric Properties

Property	Metric Value	Imperial Unit Equivalent
Outer Width × Height	200 mm × 200 mm	7.874 in × 7.874 in
Wall Thickness (t)	3.0 mm	0.118 in (approx. 11 Gauge)
Nominal Mass per Meter	~18.3 kg/m	~12.3 lbs/ft
Cross-Sectional Area (A)	23.3 cm ²	3.61 in ²
Moment of Inertia (I _x / I _y)	1480 cm ⁴	35.56 in ⁴
Section Modulus (W _{el})	148 cm ³	9.03 in ³
Radius of Gyration (i)	7.96 cm	3.13 in

2. Manufacturing Standards & Material Grades

This profile is generally manufactured in compliance with one or more of the following international regulatory frameworks:

Standard	Common Available Grades	Application Environment
EN 10210 / EN 10219	S235JRH, S275J0H, S355J2H	European structural applications, high load profiles
ASTM A500	Grade A, Grade B, Grade C	North American structural framing, welded, bolted frames
AS/NZS 1163	C250L0, C350L0	Australian/Oceania structural/mechanical load tasks
JIS G3466	STKR400, STKR490	Japanese engineering and civil structures



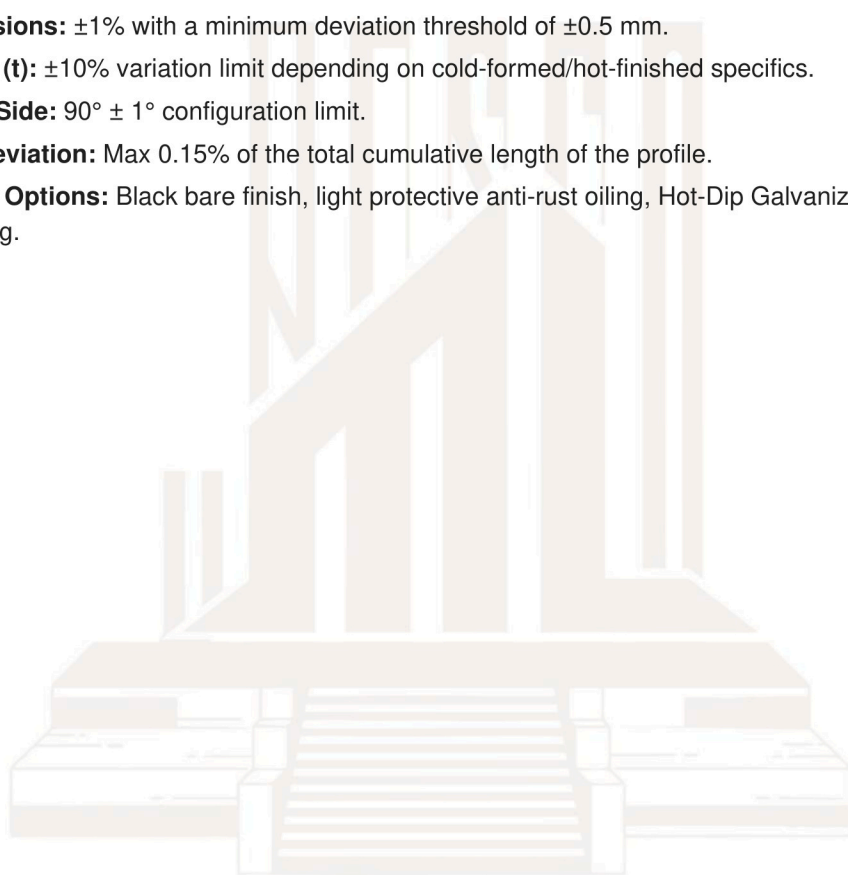
3. Mechanical & Chemical Characteristics (Typical)

Values mapped below reflect common high-strength variations (e.g., S355J2H or ASTM A500 Grade C):

Property Description	Typical Limit (Metric)	Typical Limit (Imperial)
Minimum Yield Strength (R _e)	355 MPa	51,500 psi
Ultimate Tensile Strength (R _m)	470 – 630 MPa	68,000 – 91,300 psi
Minimum Elongation (A ₅)	20%	20%
Carbon Equivalent Value (CEV max)	0.45%	0.45%

4. Standard Tolerances & Surface Customization Options

- **Outside Dimensions:** $\pm 1\%$ with a minimum deviation threshold of ± 0.5 mm.
- **Wall Thickness (t):** $\pm 10\%$ variation limit depending on cold-formed/hot-finished specifics.
- **Squareness of Side:** $90^\circ \pm 1^\circ$ configuration limit.
- **Straightness Deviation:** Max 0.15% of the total cumulative length of the profile.
- **Surface Supply Options:** Black bare finish, light protective anti-rust oiling, Hot-Dip Galvanized (HDG), or mill varnished priming.





ELEMENT POLE

1. Steel

Quality

Shaft Of Pole : Steel grade Fe 510c or Equiv.

Base Plate, Bracket : Steel Grade St 37.2

Machining

Holes required for fixing to poles are drilled. Also the fixing part to the luminaires, depending of model, All fasteners are stainless Steel AISI 304.

Finishing

After machining, welding, Grinding, sanding and making sure that all the impurities are removed, two-component primer is applied for adherence increase.

1- Hot Dipped Galvanized According to ASTM A123 or BS EN ISO 1461 or Equiv.

2- final polyurethane layer is applied for the colour, 60 micron thick., usually Matt black. For other colours, please indicate RAL number

2. Extruded Aluminum

Quality

Shaft & Bracket made from extruded aluminum , Alloy 6063 and tempering treatment is T6.

Base Plate and decorative Parts, Eutectic alloy with excellent casting properties, excellent fluidity and high resistance to hot tearing. Good machinability, excellent weldability, and high chemical resistance, EN-AC-44100 Casting, According to EN 1706 and Casting of 7.8kg/mm²

Machining

Holes required for fixing to poles are drilled. Also the fixing part to the luminaires, depending of model, All fasteners are stainless Steel AISI 304.

Finishing

After machining, welding, Grinding, sanding and making sure that all the impurities are removed, polyurethane layer is applied for the colour, 60 micron thick., usually Matt black. For other colours, please indicate RAL number



Fuse Box

PMMA or Aluminum





Light Characteristics

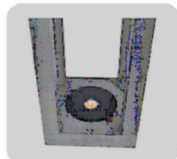
Technical Notes:

- HeatSink Aluminum Module.
- high-efficiency High Power LEDs, serial connected.
- degree of protection : IP68 IK08
- ESD protection class II

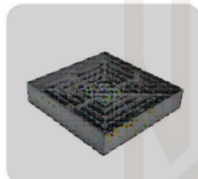
PowerAvailable : 10W,20W,30W,50W
CCT: 3000K-4000K-5000K -6000K
Efficacy: LUMILED 3030 LEDChip 127 lm/w
CREE COB LEDChip 140lm/w

Operating Life: 50,000hrs

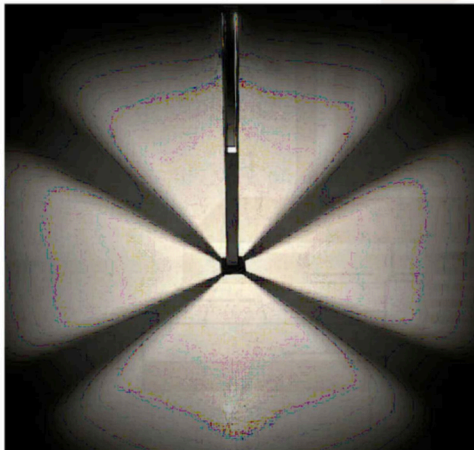
ALUMINUM LED MODULE



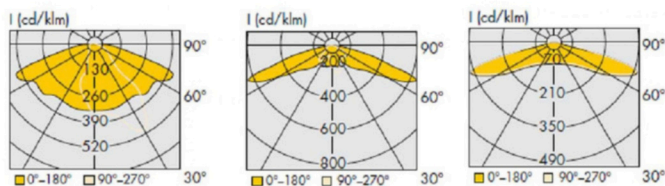
Ultrawhite glass



Die casting heatsink



Typical Light Distribution Curve:





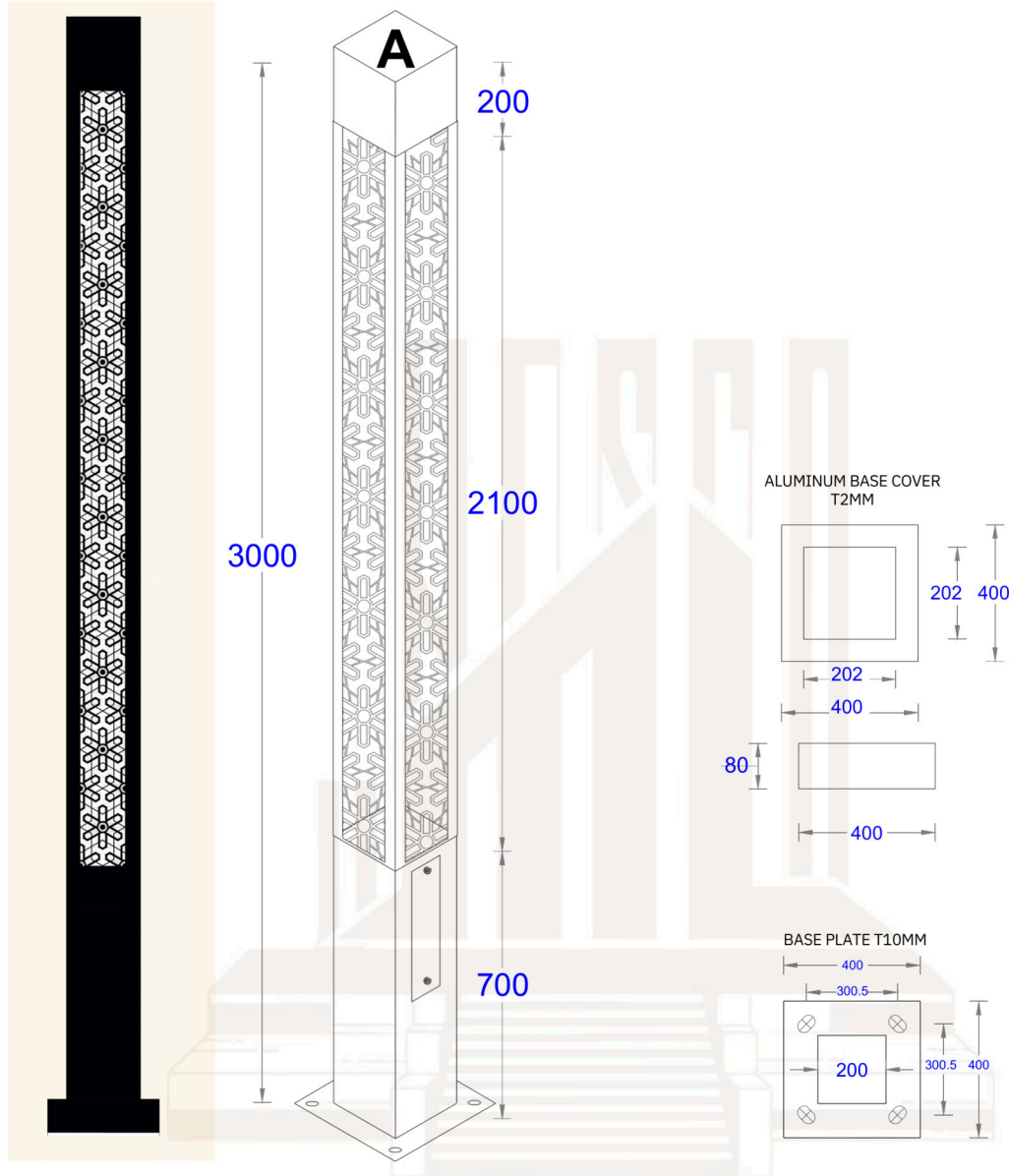
شركة الخطوط المعدنية

للمقاولات العامة

Metal Lines Co.

س.ت. 4031308794 C.R.

Drawing



Body Made From:

☐ Steel

☒ Aluminum

Finish Surface:

Acrylic Diffuser:

☐ Without

☒ 3MM ACRYLIC LEXAN

POWDER COATING RAL9017

Square Size:

☐ 200X200

☒

☐ 220 X 220

☐ 250X250

High Available:

☐ 3M2M

☒

☐ 4M

☐ 4.5M

☐ 5M

☐ 5.5M

Wattage:

☐ 10W

☐ 30W

☒ 50W

☒ 3000K

☐ 4000K

☐ 5000K

H1 & H2 & H3:

AS PER REQUEST

*ALL DIMENSIONS ARE IN MILLIMETERS

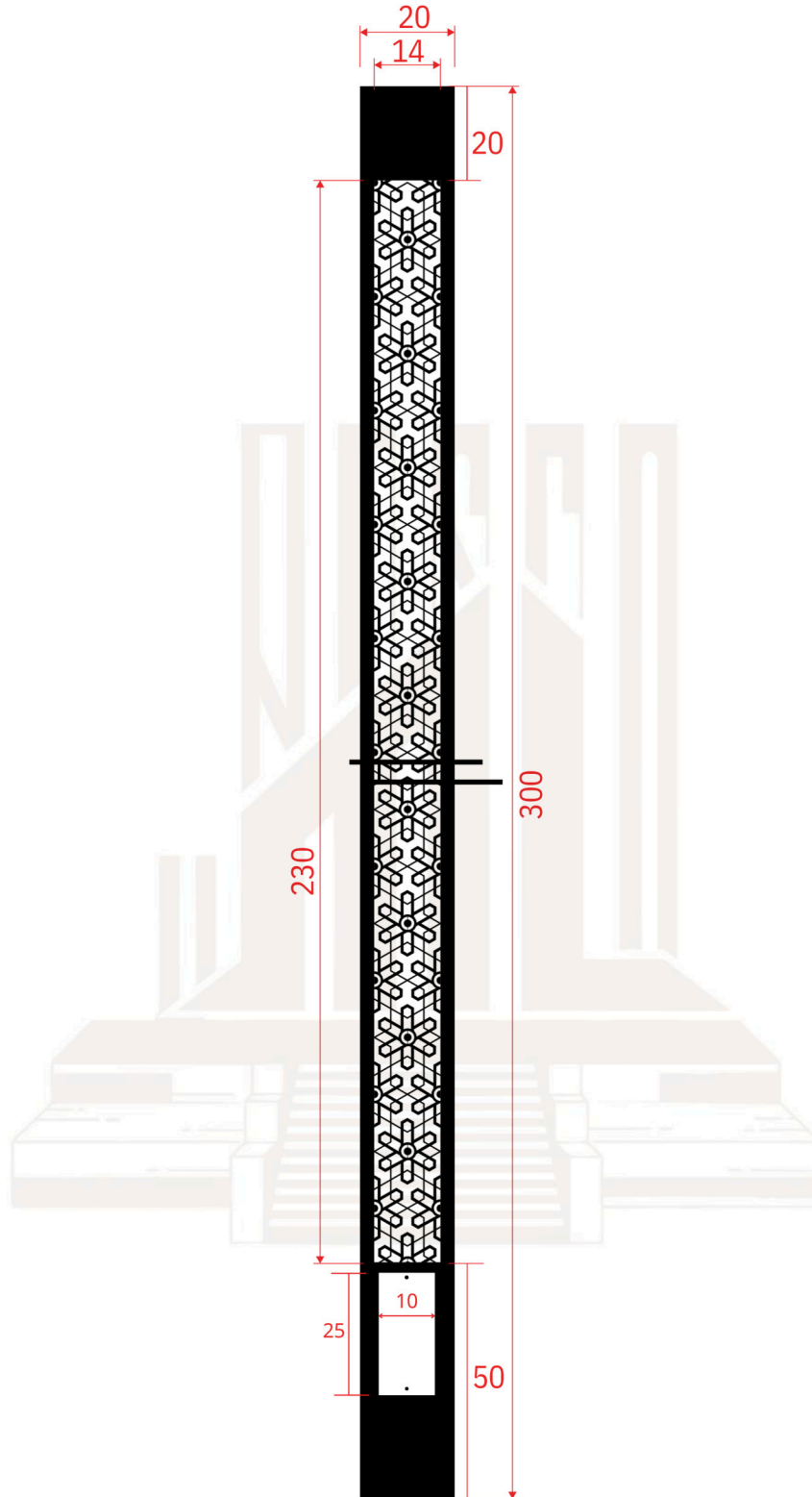


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UNILinear Flex LED Strip Light

Model: BGC401 400LM 10W 3000K L5 0612 S G

The UNILinear Flex BGC401 is a professional-grade, highly durable waterproof and dustproof LED strip light. Designed specifically for outdoor architectural decoration, cove lighting, and path guidance, it utilizes advanced anti-UV polyurethane irrigation sealing glue technology to preserve pristine color consistency and safeguard components in heavy weather environments.

Technical & Performance Specifications

Parameter	Specification Value
Color Temperature (CCT)	3000 K (Warm White, 930)
Luminous Flux	2,000 lm per 5-Meter Reel
Power Consumption	43 W total (approx. 10W/m matrix)
Input Voltage	24 V DC (Constant Voltage)
Color Rendering Index (CRI)	≥80 (High fidelity color rendering)
Ingress Protection (IP Code)	IP66 (Dust-tight, protected against powerful water jets)
Mechanical Impact Protection	IK04 (0.5 J standard protection)
Dimensions	6 mm Height × 12 mm Width × 5000 mm Length
Housing & Cover Material	UV-resistant Silicon Rubber & Stainless Steel fixations
Operating Temperature Range	-40 °C to +45 °C
Dimmability	Non-Dimmable (Constant Light Output configuration)
Cut Ability	Not cuttable (to preserve strict IP waterproofing integrity)
Lifespan	30,000 Hours (L70B50 @ 35 °C)
Connection Mode	Flying leads connection with 0.6 meter integrated cable

Key Benefits & Application Guidelines

- Advanced Weather Shielding:** Sealed with anti-UV polyurethane without color-shifting or premature aging under heavy solar exposure.
- Color Uniformity:** Tight binning checks (SDCM ≤5) ensure seamless light distribution along continuous lines without visible dark spots.
- Safe Installation:** Class III IEC safety class rating utilizes 24V low voltage loops. *Warning: Do not cut this product; altering length breaches irrigation seals and forfeits warranty coverage.*